A Zeroing Neural Network for Solving Discrete Time-varying Minimization with Different Adjustable Parameter

Haolin HE

School of Information and Communication Engineering
University of Electronic Science and Technology of China

Introduction

In recent years, researches in minimization have rising heat

The control of mechanical arms, health big data digitization, and logistics engineering

Although there are many minimization algorithms, a considerable part of them are designed to solve time-invariant problems

In reality there are still time-varying problems that need to be addressed

It is of great importance to develop minimization model to deal with time-varying problems

Among many minimization algorithms, **ZNN** is widely used due to its special structure

Introduction

Hopfield neural network

ZNN is a product of
Hopfield neural network research

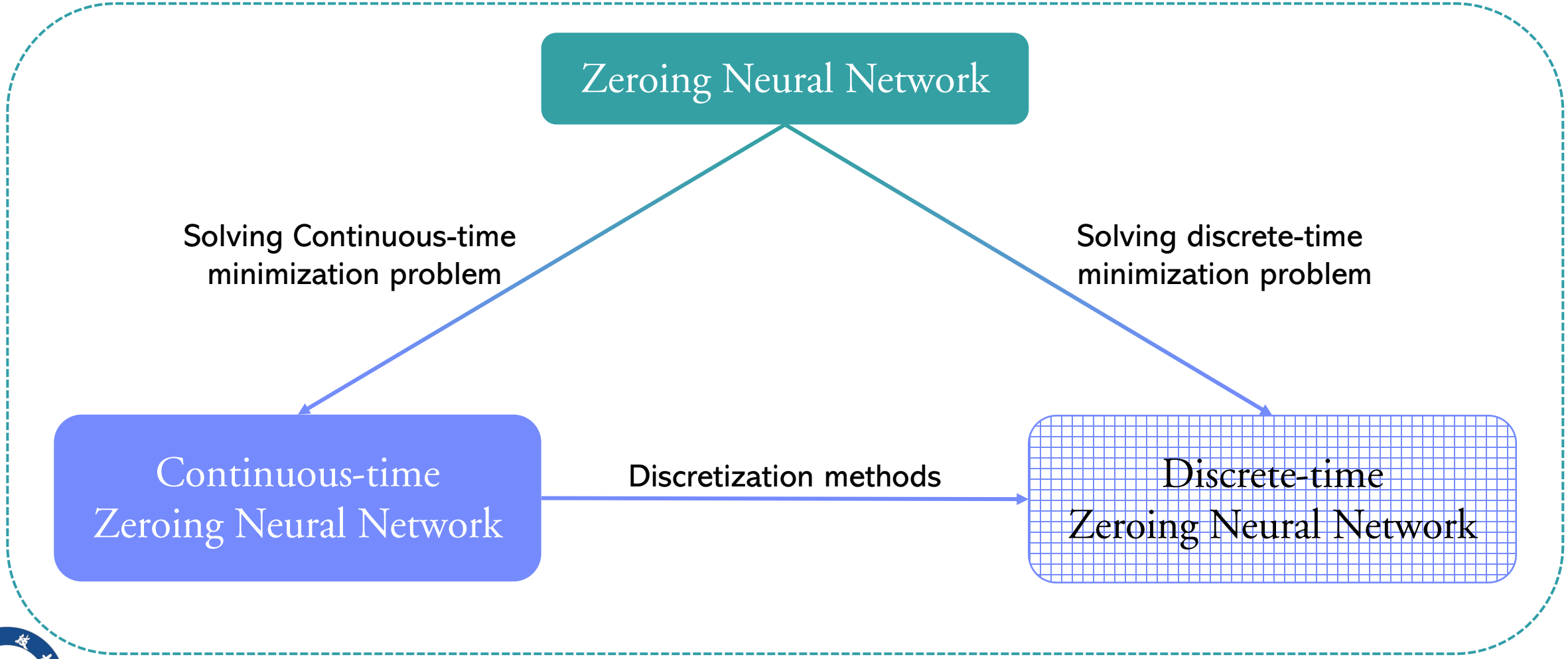
Zeroing Neural Network

Recurrent neural network (RNN)

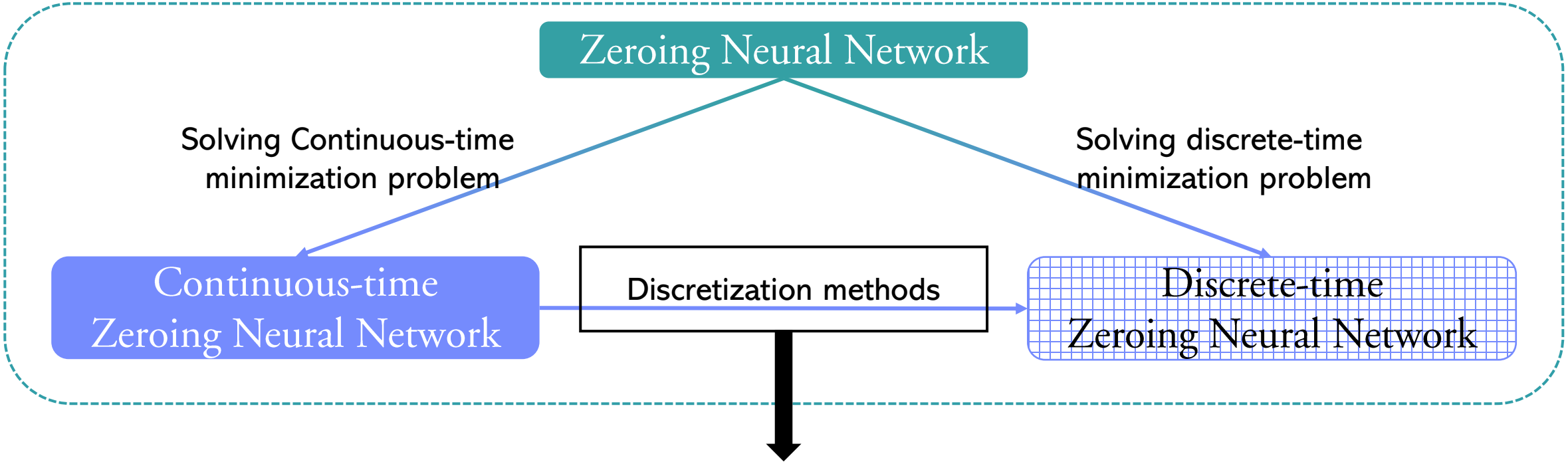
perform operations in sequence → save the above information to an intermediate state
→ comprehensively consider when processing the following information to complete context awareness

- Zeroing neural network is a special case of neural network with this structure, which has the structural characteristics of RNN and also has some independent designs.
- It is used to solve the minimization problem of time-varying signal.

Introduction



Introduction



- In this paper, to further optimize and add more flexibility to the related DTZNN, we present the general four-instant discretization formula, which has an adjustable parameter d, for the discretization process.
- These parameters on the one hand represent the flexibility of the discrete process, and on the other hand have an impact on the final result.

Problem Formulation

Before discretization, the CTZNN approach can be used to create the CTZNN model initially.

$$\min_{\mathbf{x}(t)\in\mathbb{R}^n} \boxed{\boxed{f(\boxed{\mathbf{x}(t)},t)} \in \mathbb{R}, t \in [t_0, t_f] \subseteq [0,+\infty)}$$

Gradient is required to be zero

- It is always considered as neural state.

- It is a time-varying nonlinear mapping function.
- In ZNN theory, is also considered as a time-varying signal.

$$\psi(\mathbf{x}(t),t) = \frac{\partial f(\mathbf{x}(t),t)}{\partial \mathbf{x}(t)} \in \mathbb{R}^n$$

Problem Formulation

$$\psi(\mathbf{x}(t),t) = \frac{\partial f(\mathbf{x}(t),t)}{\partial \mathbf{x}(t)} \in \mathbb{R}^n$$

Apply the ZNN design formula $\mathbf{e}(t) = \mathrm{d}\mathbf{e}(t)/\mathrm{d}t = -\gamma \mathbf{e}(t)$

$$\mathbf{e}(t) = \psi(\mathbf{x}(t),t)$$

$\gamma > 0$ is used to adjust the convergence rate of ZNN

$$H(\mathbf{x}(t),t)\dot{\mathbf{x}}(t) + \psi_t'(\mathbf{x}(t),t) = -\gamma\psi(\mathbf{x}(t),t)$$

$$\psi_t'(\mathbf{x},t) = \frac{\partial \psi(\mathbf{x}(t),t)}{\partial t} \in \mathbb{R}^n$$

$$H(\mathbf{x}(t),t) = \frac{\partial \psi(\mathbf{x}(t),t)}{\partial \mathbf{x}^{\mathrm{T}}} \in \mathbb{R}^{n\times n}$$

Problem Formulation

$$H(\mathbf{x}(t),t)\dot{\mathbf{x}}(t)+\psi_t^{'}(\mathbf{x}(t),t)=-\gamma\psi(\mathbf{x}(t),t)$$

- Switching perspectives: extracting the only derivative
- Derivative is the key of discretization

$$\dot{\mathbf{x}}(t)=-H^{-1}(\mathbf{x}(t),t)(\gamma\psi(\mathbf{x}(t),t)+\psi_t^{'}(\mathbf{x}(t),t))$$

$$\min_{\mathbf{x}(t)\in\mathbb{R}^n} f(\mathbf{x}(t),t)\in\mathbb{R} \xrightarrow[\text{has changed}]{\text{Optimization goal}} \min_{\mathbf{x}_{k+1}\in\mathbb{R}^n} f(\mathbf{x}_{k+1},t_{k+1})\in\mathbb{R},\ t=(k+1)\delta$$

$\mathbf{x}_{k+1}$ need to be generated between t_k and t_{k+1}

Proper theory for discretization and differentiation is required !

$$\dot{\mathbf{x}}_k=-H^{-1}(\mathbf{x}_k,t_k)(\gamma\psi(\mathbf{x}_k,t_k)+\psi_{t_k}^{'}(\mathbf{x}_k),t_k))$$

Problem Formulation

Lemma 1 *There is an O truncation Error* $O(\delta^2)$ *in the second-order derivative elimination (SODE) method's four-instant discretization formula. The effective domain of parameter* d *is* $d \in (-\infty, -1) \cup (1/3, +\infty)$ *for the ZNN discretized by such a formula.*

$$\dot{\mathbf{x}}_k = \frac{2d}{(3d-1)\delta}\mathbf{x}_{k+1} - \frac{3d+3}{2(3d-1)\delta}\mathbf{x}_k + \frac{2}{(3d-1)\delta}\mathbf{x}_{k-1} + \frac{-1-d}{2(3d-1)\delta}\mathbf{x}_{k-2} + O(\delta^2)$$

$$\dot{\mathbf{x}}_k = -H^{-1}(\mathbf{x}_k, t_k)(\gamma\psi(\mathbf{x}_k, t_k) + \psi'_{t_k}(\mathbf{x}_k), t_k))$$

Discrete by lemma 1

Discrete by Taylor-type differentiation formula

$$\mathbf{x}_{k+1} = \frac{(3d+3)}{4d}\mathbf{x}_k - \frac{1}{d}\mathbf{x}_{k-1} + \frac{d+1}{4d}\mathbf{x}_{k-2} - \frac{(3d-1)\delta}{2d}H^{-1}(\mathbf{x}_k, t_k)\gamma\psi(\mathbf{x}_k, t_k) - \frac{(3d-1)\delta}{2d}H^{-1}(\mathbf{x}_k, t_k)\psi'_t(\mathbf{x}_k, t_k) - O(\delta^3)$$

d is set to 1

$$\mathbf{x}_{k+1} = 1.5\mathbf{x}_k - \mathbf{x}_{k-1} + 0.5\mathbf{x}_{k-2} - H^{-1}(\mathbf{x}_k, t_k)\gamma\psi(\mathbf{x}_k, t_k) - H^{-1}(\mathbf{x}_k, t_k)\psi'_t(\mathbf{x}_k, t_k) - O(\delta^3)$$

Experiments and Verifications

- Design of time-varying mapping functions

$$f_1(\mathbf{x}_k,t_k)=(3+\sin(t_k))x_1(t_k)^2+2x_2(t_k)^2+(x_3(t_k)-e^{-t_k})^2+(x_4(t_k)+e^{-t_k})^2+(x_1(t_k)+\sin(t_k))x_3(t_k)+0.1(t_k-1)x_3(t_k)x_4(t_k)$$

$$f_2(\mathbf{x}_k,t_k)=(2+\cos(t_k))x_1(t_k)^2+x_2(t_k)^2+3x_3(t_k)^2+x_4(t_k)^2-\cos(t_k)x_1(t_k)x_3(t_k)-\exp(-t_k)x_2(t_k)+3x_3(t_k).$$

- Design of time-varying mapping functions

The initial vector $\mathbf{x}_0 \in [-0.25, 0.25]^4$ is a vector which is generated randomly. Besides, the following two vectors $\mathbf{x}_1$ and $\mathbf{x}_2$ are generated by $\mathbf{x}_0$ according to the Newton iteration, for its structure is the most basic.

$$\mathbf{x}_{k+1}=\mathbf{x}_k-H^{-1}(\mathbf{x}_k,t_k)\psi(\mathbf{x}_k,t_k)$$

Experiments and Verifications

- Design of experiment 1

The first experiment is designed to analyze the evolution of residual error. In this experiment, we study the change of error with the change of δ and prove the order of residual error to be $O(\delta^3)$.

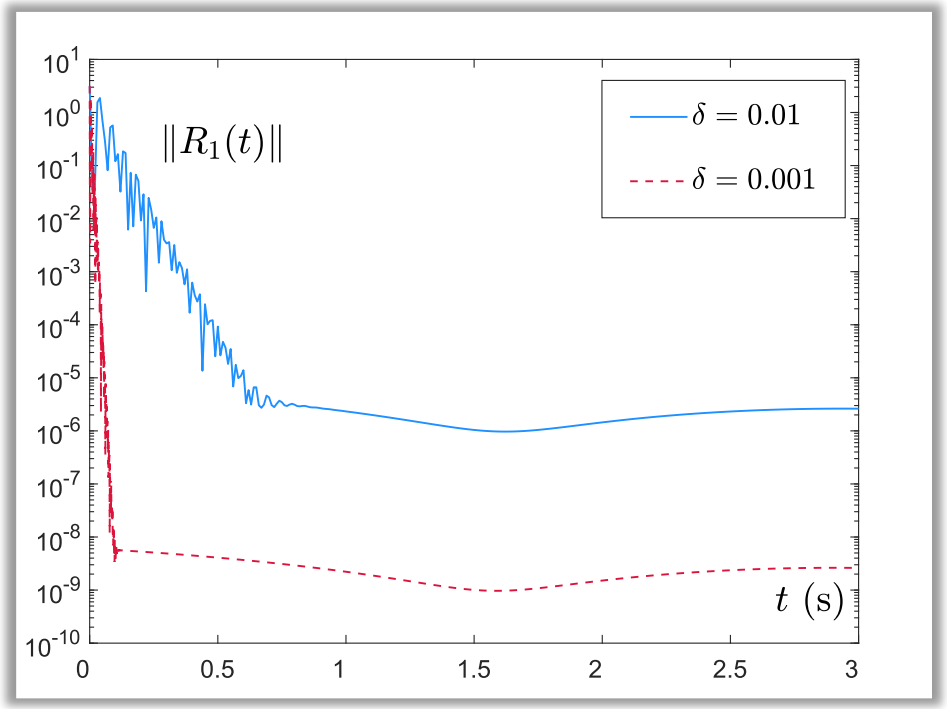
In DTZNN, $h = \gamma\delta > 0$ is often used to represent the iteration step size. In this experiment, to control the variable, d is set to be constant 1 and h is set to be constant 0.3.

	d	h	δ
constant	✓	✓	✗
value	1	0.3	0.01 & 0.001

Table 1: State of variables in experiment 1

Experiments and Verifications

- Results of experiment 1



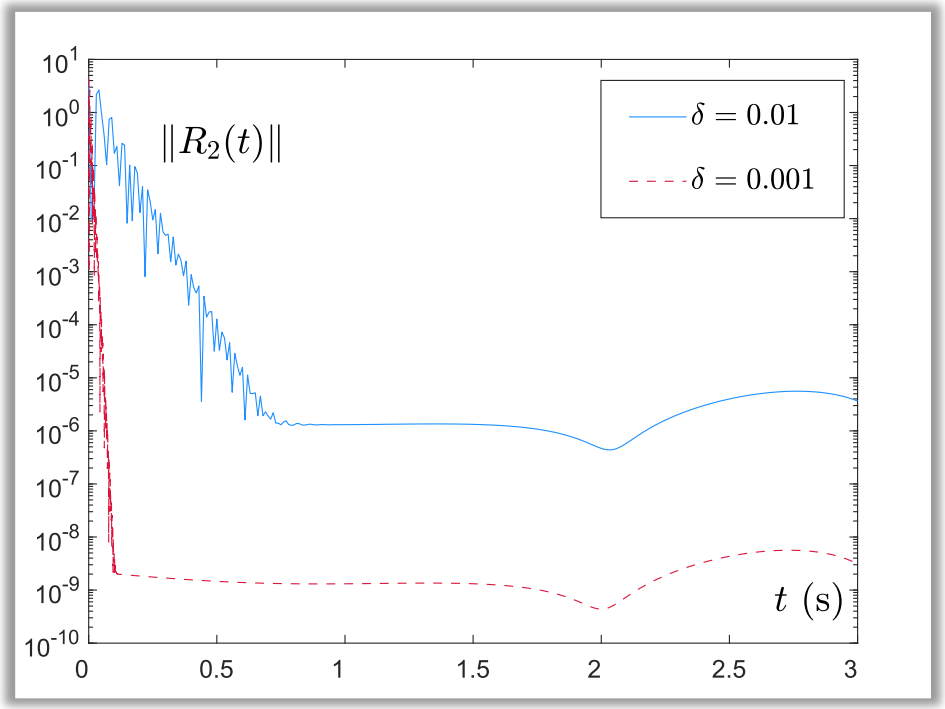


Fig. 1: Semilog plot of residual errors of two time-varying signal $f_1(\mathbf{x}_k, t_k)$ and $f_2(\mathbf{x}_k, t_k)$, where $t = (k+1)\delta$ and $h = 0.3$

Experiments and Verifications

- Design of experiment 2

The second experiment is aimed to study the variation of residual error with d.

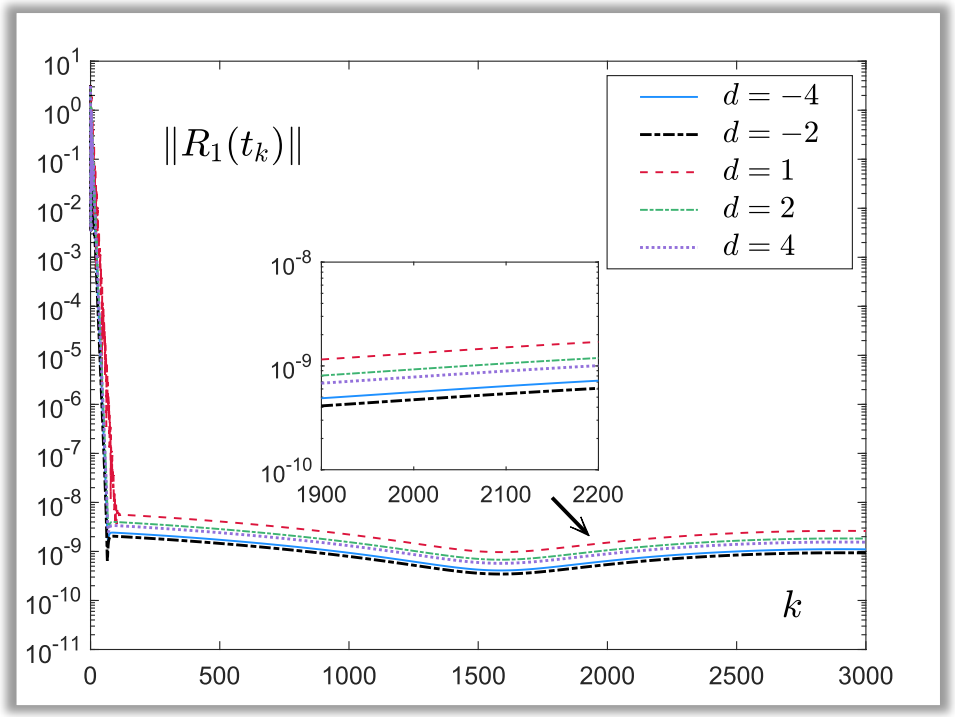
In contrast to the first experiment, to control variables, δ is fixed to be constant at 0.001 and h is set to be constant at 0.3 in the second experiment.

	d	h	δ
constant	✗	✓	✓
value	-4, -2, 1, 2, 4	0.3	0.001

Table 1: State of variables in experiment 2

Experiments and Verifications

- Results of experiment 2



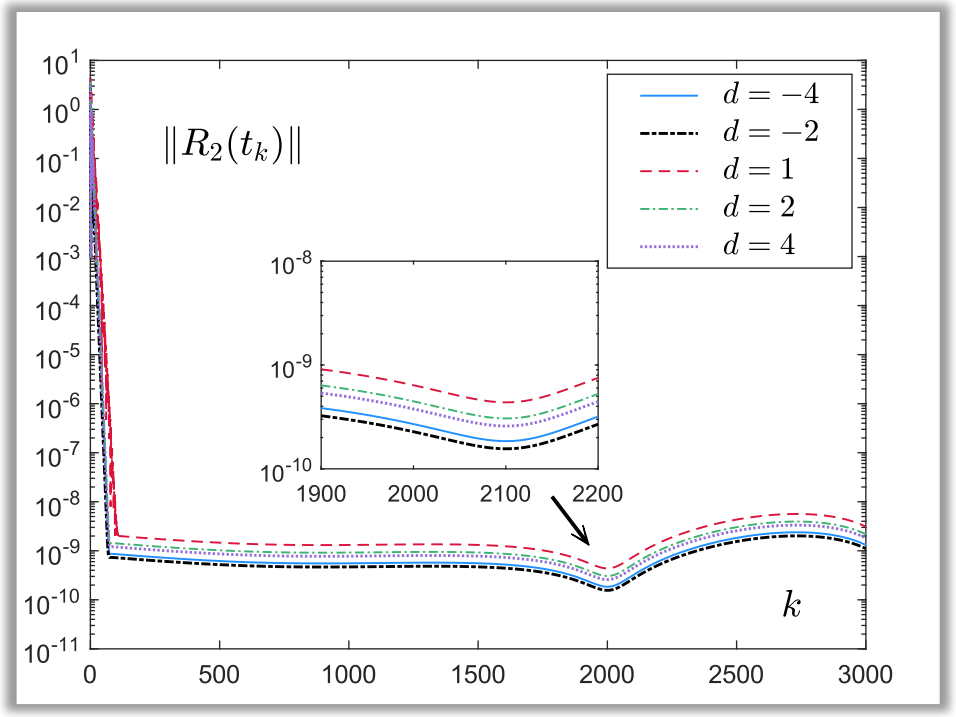


Fig. 2: Semilog plot of residual errors of two time-varying signal $f_1(\mathbf{x}_k, t_k)$ and $f_2(\mathbf{x}_k, t_k)$, where $\delta = 0.001$ and $h = 0.3$.

Conclusion

- The above four properties demonstrate the effectiveness of our proposed method, and the choice of d value will indeed have a significant impact on the final effect.

- Users can choose d in practical application according to the characteristics of the algorithm, which takes into account the flexibility and effectiveness of the algorithm.

THANKS FOR WATCHING !

Haolin HE

School of Information and Communication Engineering
University of Electronic Science and Technology of China